Monthly Intelligence Report

Edition 001 — The Ontario–Quebec Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED
24,986
996 EHV · 6,902 HV

GRID LINES MAPPED
~12,400
~105,000 km coverage

MC SIMULATIONS
78.9 M
10,000 per substation

PUBLIC DATA SOURCES
30+
95 variables · zero proprietary

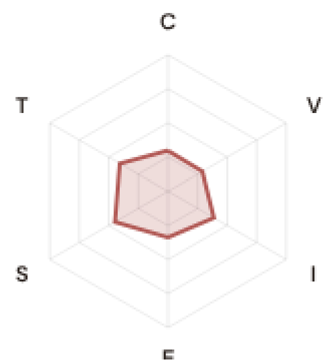
The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 78.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 24,986 substations and ~12,400 grid lines spanning ~105,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Canada's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Ontario, Ontario · 115 kV

0.371	Medium	0.270	44th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.297 / 0.30 (99%)
- **I** Infrastructure: 0.388 / 0.25 (155%)
- **S** Saturation: 0.448 / 0.20 (224%)
- **V** Voltage: 0.289 / 0.10 (289%)
- **E** Economic: 0.336 / 0.10 (336%)
- **T** Transition: 0.407 / 0.05 (814%)

MODIFIERS

- **R3 Consequence:** 0.945 ↓ dampens
- **R4 Graph Criticality:** 1.036 ↑ amplifies
- **R6a Restoration:** 1.042 ↑ amplifies
- **R6b Seismic:** 1.000 → neutral
- **R7 Digital Readiness:** 1.008 → neutral

This month's spotlight — automatically selected for April 2026 — is in Ontario, Ontario. With an R_final of 0.371 (Medium band, 44th fleet percentile), its risk profile is dominated by the T (Transition) component at 814% of its weight ceiling, followed by E (Economic) at 336%. Modifiers collectively shift R_base by 2.8%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Canada Numérique / Canada 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

1303

High/Critical + R7 < median

AVG R7 MODIFIER

1.0100

Range [0.99, 1.05]

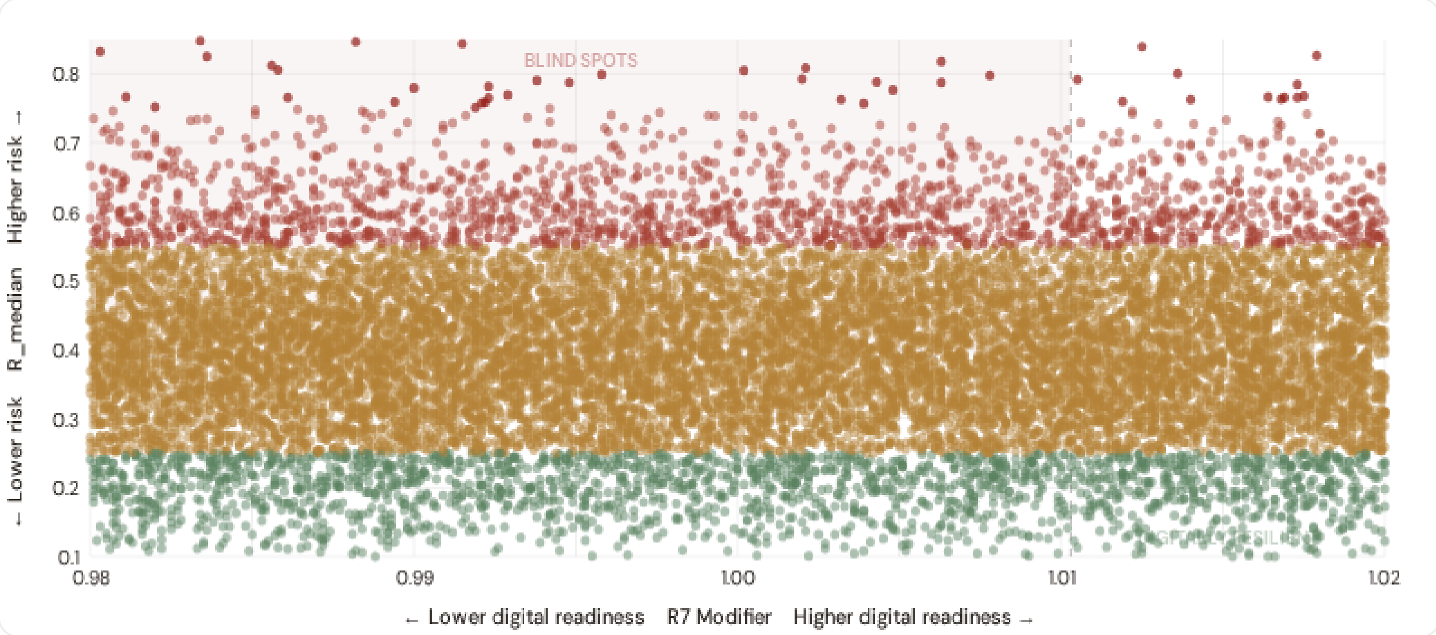
HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **1303 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0103). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Ontario (412), Manitoba (400), British Columbia (181)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the southern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure -- highest economic loss per interruption

6,249 substations (25.0%) High/Critical: **686** (11.0%) Avg R: **0.392** Avg E: **0.446** / 0.10

Low 807 Med 4756 High 664 Crit 22

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: Est. medium-high VoLL

Diversified industry, logistics, processing -- significant continuity requirements

6,262 substations (25.1%) High/Critical: **671** (10.7%) Avg R: **0.392** Avg E: **0.464** / 0.10

Low 776 Med 4815 High 647 Crit 24

🏢 Commercial / SME-Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

6,230 substations (24.9%) High/Critical: **649** (10.4%) Avg R: **0.391** Avg E: **0.481** / 0.10

Low 820 Med 4761 High 630 Crit 19

🌾 Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

6,245 substations (25.0%) High/Critical: **640** (10.2%) Avg R: **0.391** Avg E: **0.499** / 0.10

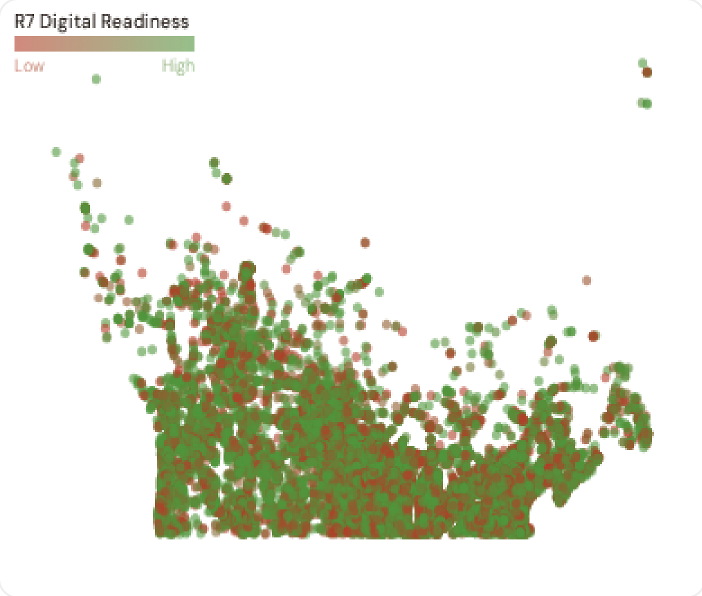
Low 859 Med 4746 High 619 Crit 21

Value of Lost Load (VoLL) context: CER's 2024 estimate for Canada places average VoLL at C$17/kWh for industrial customers and C$4.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Ontario (2015), Manitoba (1586), British Columbia (998)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of C$20–40/kWh, compared to C$1–4/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 8.0%, but this masks sharp regional variation: Remote and northern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Census Division Digital Readiness Ranking

Census Divisions ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify census divisions combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 PROVINCES

Longer bar = higher R7 = better digital infrastructure.

1	Nunavut ⚠ high R	1.0091
2	New Brunswick ⚠ high R	1.0091
3	Quebec	1.0094
4	Newfoundland and Labrador ⚠ high R	1.0096
5	Alberta	1.0098
6	Ontario	1.0100
7	Manitoba ⚠ high R	1.0101
8	Saskatchewan ⚠ high R	1.0102
9	Nova Scotia ⚠ high R	1.0105
10	British Columbia	1.0106
11	Yukon ⚠ high R	1.0124
12	Northwest Territories ⚠ high R	1.0171

Census Division-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Nunavut** has the lowest average R7 (1.0091), while **Northwest Territories** leads at 1.0171 — a spread of 0.0080 across the R7 range. **5 census divisions** combine below-median digital readiness with above-median grid risk, making them priority targets for Canada 2030 / Canada CIB / IRP digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (April 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0100, median = 1.0103; 0 substations classified HIGH cyber-exposure; 1303 blind-spot substations (High/Critical risk + below-median R7); 33 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging census divisions where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **32 verified public data sources** feeding 95 variables across 5 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

32

95 variables

WEEKLY / MONTHLY

20

High-frequency feeds

QUARTERLY / ANNUAL

12

Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

Data Source Registry — April 2026

Loading source registry...

C.1 Update Frequency Timeline

Loading...

C.2 Fleet Score Snapshot

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C.3 Changelog — v3.4 → v4.0.2

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SECTION D — MONTHLY DEEP-DIVE

The Ontario–Quebec Corridor: Where Hydroelectric Dominance Meets Grid Modernization

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Ontario–Quebec corridor · **002** DER saturation hotspots (Alberta/Saskatchewan) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan-Rural economic impact disparity · **005** Nuclear fleet dependency and transition risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Census Division Grid Health Cards · **010** Academic validation and peer feedback · **011** North American replication feasibility · **012** Year in review.

D.1 Why the Ontario–Quebec, Why Now

ONTARIO–QUEBEC SUBSTATIONS

HIGH BAND

1015 + 26

CORRIDOR MEDIAN R

0.379

11402

1838 EHV · 9564 HV

9.1% of corridor

Fleet median: 0.389

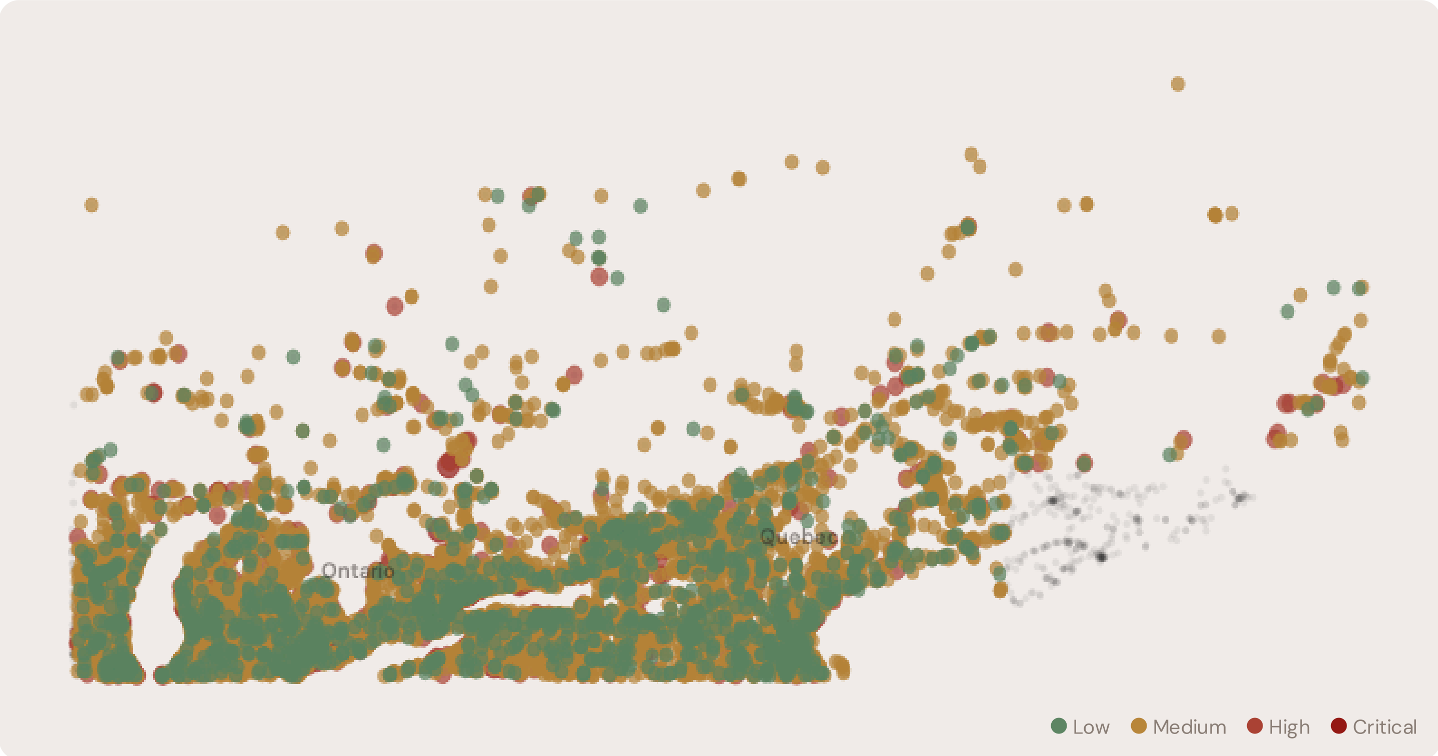
WORST CENSUS DIVISION

Ontario

Avg R: 0.389 · 8271 substations

The Ontario–Quebec hosts **11402 substations** across six census divisions, making it one of the most energy-critical corridors in Europe. However, its risk profile is disproportionately concentrated in the upper bands: **1015 substations (8.9%)** are classified High and **26** Critical, with only **1658** in the Low band. 47% of Ontario–Quebec substations score above the national fleet median of 0.389. The corridor median R of 0.379 reflects the tension between nuclear baseload inflexibility, rapidly growing renewable variability, and aging grid infrastructure built for 1970s load profiles. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under Canada 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
	Ontario	0.901	Critical	0.681	0.931	1.406	1.075	0.735	1.440	
	Ontario	0.818	Critical	0.867	0.850	1.091	1.067	1.016	0.979	
	Ontario	0.814	Critical	0.981	0.458	0.903	0.979	0.900	0.965	
East Jordan Junction	Ontario	0.812	Critical	0.658	0.469	1.270	0.833	0.941	1.419	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Herridge Lake D.S.	Ontario	0.808	Critical	0.808	0.740	0.723	0.737	0.554	0.799	
Taylorville	Ontario	0.806	Critical	0.614	0.751	1.243	1.049	0.792	0.790	
	Ontario	0.802	Critical	0.868	0.645	0.990	1.070	1.107	0.887	
	Ontario	0.792	Critical	0.935	0.736	1.131	1.081	0.654	1.419	
	Ontario	0.787	Critical	0.770	0.723	1.075	0.747	0.656	0.764	
	Quebec	0.787	Critical	0.868	0.659	1.021	0.618	0.788	0.974	

… 11392 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — ONTARIO–QUEBEC VS FLEET

C — Continuity 0.386 vs 0.394 (-2%)

I — Infrastructure 0.514 vs 0.524 (-2%)

S — Saturation 0.387 vs 0.394 (-2%)

V — Voltage 0.310 vs 0.316 (-2%)

E — Economic 0.463 vs 0.473 (-2%)

T — Transition 0.515 vs 0.525 (-2%)

MODIFIER AVERAGES — ONTARIO–QUEBEC VS FLEET

R3 Consequence	0.954	fleet 0.955	↓
R4 Graph Crit.	0.951	fleet 0.950	↓
R6a Restoration	0.975	fleet 0.975	↓
R6b Seismic	1.004	fleet 1.010	→
R7 Digital Read.	1.010	fleet 1.010	→

The dominant risk driver across the Ontario–Quebec corridor is **T (Transition)**, averaging 0.515 against a fleet average of 0.525 — occupying 1030% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.463 vs fleet 0.473 (463% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.954 (fleet: 0.955), reflecting the socio-economic fragility of The Ontario–Quebec's industrial-oriented, manufacturing-dependent economy with significant hydroelectric and nuclear bases. The R6b seismic modifier averages 1.004, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 Census Division Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Ontario 8271 subs · avg R = 0.389 · H:825 C:24 M:6289

Quebec 3131 subs · avg R = 0.364 · H:190 C:2 M:2414

The province-level breakdown reveals significant intra-regional variation. **Ontario** leads with a mean R of 0.389 across 8271 substations, while **Quebec** scores 0.364 — a spread of 0.025 within a single region. This disparity underscores why regional-level metrics (as used by NERC and IEEE) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the Ontario–Quebec is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

197 Ontario–Quebec substations score R ≥ 0.650, placing them in the upper quartile of national risk. For IRP (Integrated Resource Planning) investment prioritisation, this analysis identifies the Ontario corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

North American Context Card

How this works: Each edition includes one North American Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Ontario–Quebec corridor analysis hinges on Canada's continuity performance relative to international peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs US (003), CMIP6 climate hazard vs Northern US peers (004), etc. The underlying data updates annually with the NERC Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected International Peers

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.

🇩🇪 Germany	11.7
🇨🇭 Switzerland	~15
🇳🇱 Netherlands	23.9
🇨🇦 Canada (national)	**~36**
↳ Urban	**~22**
↳ Rural	**~58**
🇬🇧 UK	~40
🇮🇹 Italy	~43
🇪🇸 Spain	~48
🇭🇺 Hungary	~57
🇵🇱 Poland	~95
🇷🇴 Romania	~290

Source: NERC Reliability Report (2024); CER TIQE (2023); NEB / CER (2024). Canada sub-categories from CER territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with NERC BR release.

This month's key insight: The Ontario–Quebec corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Canada's rural territory classification. The Ontario–Quebec's average C of NaN is **NaN× the fleet average** of 0.394. In comparative terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~90–180 min/customer/year) than to Canada's own national average (~36 min). The SSI reveals what aggregate NERC statistics conceal: that Canada's mid-tier international position is not a national condition but a statistical average of top-tier provincial performance and remote-territory-level rural performance — and the Ontario–Quebec corridor concentrates the worst of the latter.

Regional Risk Ranking — All 13 Canadian Provinces

Where does the Ontario–Quebec sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

#1 Nunavut — 13 subs · avg R = 0.676 · H:8 C:4

#2 Northwest Territories — 1 subs · avg R = 0.663 · H:1

#3 Yukon — 68 subs · avg R = 0.559 · H:21 C:11

#4 Newfoundland and Labrador — 166 subs · avg R = 0.463 · H:37 C:3

#5 Nova Scotia 162 subs · avg R = 0.435 · H:25 C:2

#6 Saskatchewan 1611 subs · avg R = 0.413 · H:204 C:9

#7 New Brunswick 198 subs · avg R = 0.413 · H:30 C:1

#8 Manitoba 6269 subs · avg R = 0.407 · H:783 C:22

#9 Ontario 8271 subs · avg R = 0.389 · H:825 C:24

#10 British Columbia 3927 subs · avg R = 0.383 · H:365 C:8

#11 Alberta 1169 subs · avg R = 0.368 · H:71

#12 Quebec 3131 subs · avg R = 0.364 · H:190 C:2

The Ontario–Quebec ranks **#0 of 13 provinces** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Nunavut, Northwest Territories, Yukon, while the three most resilient are Quebec, Alberta, British Columbia. 8 of 13 provinces score above the fleet average of 0.391. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across Canada. It is precisely this substation-level granularity that positions the SSI as a tool for IRP / Canada 2030 / Canada CIB investment prioritisation: not treating entire provinces as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the Hydro One/IESO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	**Layer 3 — Neural Net**	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	Hydro One/IESO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a C$13M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against CER (Commission de Régulation de l'Énergie)'s published TIQE data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall visualisation: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Canadian training set comprises 24,986 substations with complete 95-feature vectors. Average CI_width of 0.275 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 3262 Low, 19078 Medium, 2560 High, 86 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The BC Coastal Grid

Deep-dive into British Columbia's grid risk profile — substation map, component analysis, province breakdown. North American Context Card: Mediterranean thermal stress trajectory vs Atlantic peers.

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Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Canadian utility, CER, IESO, Hydro One, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 24,986 substations · ~12,400 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5
